

Motor Datasheet

Operating Point: Continuous | Number of Turns per Tooth: 75.0

Geometry

Parameter	Value	Unit
Machine Outer Diameter	59.0	mm
Machine Inner Diameter	18.0	mm
Machine Thickness	19.4	mm
Airgap	1.0	mm

Mass Properties (active elements only)

Parameter	Value	Unit
Machine Mass	286.1	g
Rotor Mass	99.4	g
Stator Mass	186.6	g
Rotor Moment of Inertia	449.8	g cm2

Electrical Properties

Parameter	Value	Unit
Supply Current (RMS)	2.9	A
Phase Resistance	2045.9	mOhm
D-Axis Inductance (Ld)	3068.9	μH
Q-Axis Inductance (Lq)	3135.8	μH
PM Flux Linkage	36.5	mWb
Frequency at operating point (electrical)	8.3	Hz
Winding Configuration	wye (star)	-

Performance

Parameter	Value	Unit
Torque (Continuous)	1126.6	mNm
Speed	100.0	rpm
BackEMF Constant (Ke)	20.1	mV/rpm
Motor Constant (Km, only resistive losses)	150.5	mNm/sqrt(W)
Torque Constant (Kt, no-load)	276.7	mNm/A_peak
Saturation	4.7	%

Losses

Parameter	Value	Unit
Total Losses	51.5	W
Resistive Losses	50.9	W

Configuration

Parameter	Value	Unit
Pole Pairs	5	-
Number of Turns per Tooth	75.0	-
Number of Phase Turns	300.0	-